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Task 1 -RACI Table

Activity/Task	Customer Analytics	Network Engineering	CRM / Retention	Call Center	Data Engineering	Management
Data Extraction & ETL	C	I	I	I	R	A
Defining KPIs Logic	R	C	C	C	I	A
Dashboard Development	R	I	I	I	C	A
Network Infrastructure Fixes	I	R	I	I	I	A
Customer Retention Campaigns	C	I	R	I	I	A
Handling Customer Complaints	I	I	I	R	I	A

Task 2- Clarification Questions

1- **Churn Definition Nuance:** Does the current definition of 'Churn' differentiate between **Voluntary Churn** (customer cancelling the contract) and **Involuntary Churn** (service cut off due to non-payment)? This distinction is crucial because the retention strategy for each group is completely different (Marketing vs. Finance).

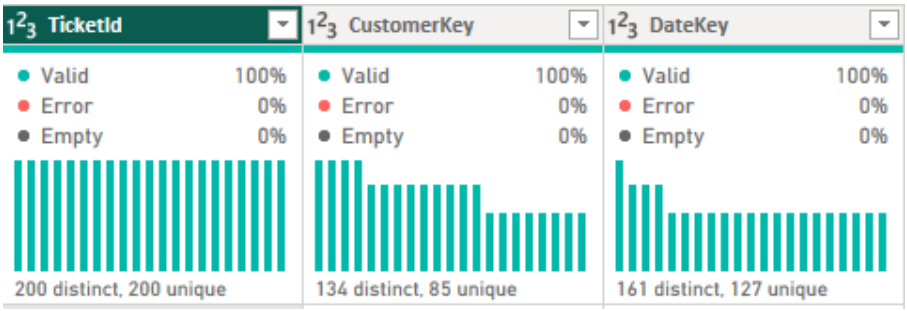
2- **High-Value Logic:** Currently, 'High-Value Customers' are identified solely based on their plan type. Should we incorporate **Customer Lifetime Value (CLV)** factors, such as tenure duration or average monthly spend on add-ons, to define this segment more accurately? A long-term customer on a medium plan might be more valuable than a new customer on a premium plan.

Task 3- Data Quality Tasks

1. How you validate **Completeness** of usage, network KPIs, churn, and tickets.

I validated completeness through a three-step process:

- **Column Quality Check:** Used Power Query's 'Column Quality' feature to ensure zero null values in primary keys (TicketID, DateKey, CellKey) across all fact tables.



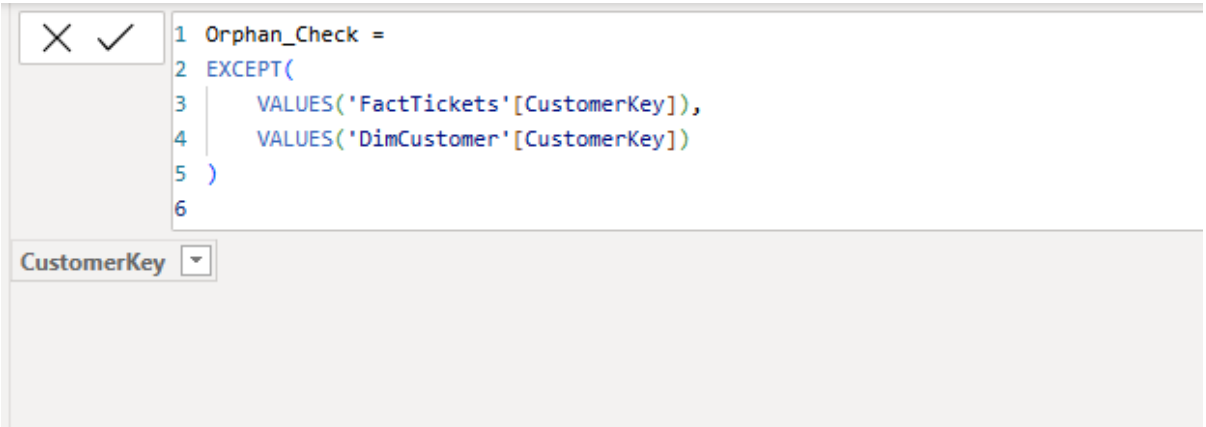
- **Temporal Continuity Analysis:** Visualized the daily record counts for FactUsageDaily and FactNetworkHourly over time to identify any missing dates or unexplained dips in data logging.

  Data Imputation (Major Finding): Identified a completeness issue in the source data where the PlanKey was missing from usage logs. I validated this gap by checking the schema and resolved it by implementing a 'Moving Average' logic to infer the missing plan information, ensuring the dataset was complete for segmentation analysis.

2. How you validate **Consistency of Customer IDs** across systems.

I validated Referental Integrity using DAX set operations to ensure no orphan records exist:

- **Orphan Check:** I created a validation table using the EXCEPT() function in DAX to compare the list of CustomerKey in Fact tables (Usage, Tickets) against the DimCustomer table.

  - *Formula used:* EXCEPT(VALUES(FactUsage[CustomerKey]), VALUES(DimCustomer[CustomerKey])).

  Result: The check returned an empty table, confirming that 100% of transaction records are linked to valid, existing customers in the master data, ensuring full consistency across the CRM and Usage systems.